BILINGUAL EDITION / ENGLISH WITH TRADITIONAL CHINESE KEYWORDS

Strawberry DNA extraction

Student worksheet · Teacher-supervised practical（實驗） · Record your own evidence.

NAME	CLASS	DATE
_____	_____	_____

Learning objectives（學習目標）

By the end of the practical（實驗）, I can…

01	Explain how membrane disruption and precipitation（沉澱） help make DNA-rich fibres（富含 DNA 的纖維） visible.
02	Describe what mashing（搗碎）, detergent（清潔劑）, salt（鹽）, filtration（過濾） and cold ethanol（冷乙醇） do.
03	Follow the method（方法） safely and record observations without claiming the crude extract is pure DNA.

Risk assessments（風險評估） & controls — read before the method（方法） and observation record（觀察紀錄）

Check the control measures with your teacher before handling any material.

MATERIAL / HAZARD（危害）	RISK（風險）	CONTROL MEASURE（控制措施）
Cold ethanol（冷乙醇） — flammable（易燃） liquid / vapour	Vapour may irritate; flames can ignite vapour.	Teacher-directed use only. Keep capped, cold and away from flames or other ignition sources（火源）.
Detergent（清潔劑） — eye / skin irritant（刺激物）	A splash can irritate eyes or skin.	Wear goggles（護目鏡）. Avoid contact; rinse splashes with plenty of water（水） and tell the teacher.
Glass and spills	Breakage can cut; liquid can make floors slippery.	Check glassware. Report breakage and clean spills promptly using the class procedure.

[ ] Goggles（護目鏡） on	[ ] Bench clear	[ ] Teacher's instructions checked

Before you begin: Which control measure（控制措施） will you check first, and why?

01 / BEFORE THE PRACTICAL（實驗）

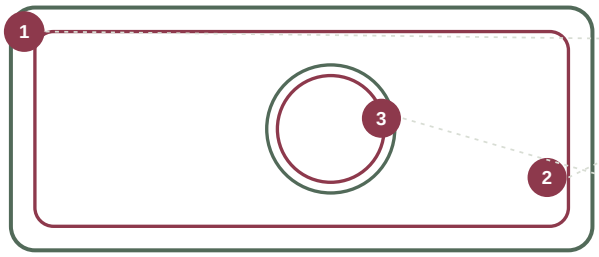
Before the practical（實驗）

Think first; return to your predictions after the experiment.

1.1 Cultivated garden strawberries are octoploid（八倍體）(8n). In your own words, what does 8n tell you about chromosome sets（染色體組）? Does ploidy（倍性） alone tell you the total DNA mass in a cell? Explain.

1.2 Use the word bank（詞彙表） to label the numbered barriers in the simplified cell sketch.

WORD BANK（詞彙表）: cell wall（細胞壁） / cell membrane（細胞膜） / nuclear envelope（核膜）



1. _____
2. _____
3. _____

SIMPLIFIED CELL SKETCH - NOT TO SCALE

1.3 Before adding the cold liquid, predict where the pale fibres（纖維） might appear and describe what you expect to see.

02 / METHOD IN PICTURES（圖解方法）

Method in pictures（圖解方法）

Follow the steps in order. Keep the filtrate（濾液）and ethanol（乙醇）layers separate.

PREPARE THE BUFFER（緩衝液）	GET THE EQUIPMENT READY
Mix 90 cm³ water（水）+ 10 cm³ detergent（清潔劑）+ 3 g salt（鹽）. Stir until the salt（鹽）dissolves. Add 10 cm³ of prepared buffer（緩衝液）to the mashed fruit.	One hulled strawberry; sealable bag; gauze（紗布）-lined funnel; clean tube; clean glass rod（玻璃棒）; goggles（護目鏡）; ice-cold ethanol（冰冷乙醇）.



01-02 / MASH（搗碎）FRUIT + BUFFER（緩衝液）

Put one hulled strawberry in a sealable bag and mash（搗碎）gently for 2 minutes. Stir the buffer（緩衝液）until the salt（鹽）dissolves, then add 10 cm³ to the fruit.

[ ] DONE



03 / FILTER（過濾）THE MIXTURE

Pour slowly through gauze（紗布）in a funnel into a clean tube. Collect the filtrate（濾液）; leave most visible pulp（果肉）on the gauze（紗布）.

[ ] DONE



04 / LAYER COLD ETHANOL（冷乙醇）

Measure 5 cm³ filtrate（濾液）and an equal 5 cm³ volume of ice-cold ethanol（冰冷乙醇）. Run ethanol（乙醇）slowly down the tube wall; do not shake or stir.

[ ] DONE



05 / SPOOL（纏繞收集）THE FIBRES（纖維）

Touch a clean glass rod（玻璃棒）to the boundary（交界面）between liquid layers. Turn slowly in one direction to collect any pale, stringy material.

[ ] DONE

WHAT TO LOOK FOR / Pale, stringy DNA-rich material may appear at the boundary（交界面）after ice-cold ethanol（冰冷乙醇）meets the filtrate（濾液）. The extract is crude, not pure DNA.

03 / OBSERVATION RECORD（觀察紀錄）

Observation record（觀察紀錄）

Describe what you see before explaining why it happened. Sketch or write; use the full space.

STAGE	WHAT I SEE / SKETCH OR DESCRIBE
After mashing（搗碎） and adding buffer（緩衝液）	
After filtration（過濾）: liquid and pulp（果肉） retained（被截留） on the gauze（紗布）	
Where ethanol（乙醇） meets the filtrate（濾液）	
After spooling（纏繞收集）: colour, texture and location	

3.1 Which observation（觀察） is clearest or most useful as evidence? Describe it without explaining it yet.

3.2 What can the appearance of fibres（纖維） show—and what cannot it prove about the crude extract?

04 / EXPLAIN AND EVALUATE

Explain and evaluate

Use evidence from your sample. A visible result is an observation（觀察）, not proof of purity.

WHAT DOES EACH STEP OR MATERIAL DO? Write an explanation in your own words.

STEP / MATERIAL	MY EXPLANATION
Mashing（搗碎）	
Detergent（清潔劑）	
Salt（鹽）	
Filtration（過濾）	
Ice-cold ethanol（冰冷乙醇）	
Spooling（纏繞收集） with a glass rod（玻璃棒）	

FILTRATION（過濾） CHECK / Tick one destination for each item.

MATERIAL	RETAINED（被截留） BY GAUZE（紗布）	IN THE FILTRATE（濾液）
Coarse pulp（果肉） and large cell-wall fragments（細胞壁碎片）	[ ]	[ ]
Unmashed strawberry tissue	[ ]	[ ]
Soluble（可溶的） DNA	[ ]	[ ]
Water（水）, salt（鹽） ions and detergent（清潔劑） molecules	[ ]	[ ]

4.1 Why is salt（鹽） added to the buffer（緩衝液）? Include DNA's charge in your explanation.

4.2 Why use ice-cold ethanol（冰冷乙醇） and add it gently without mixing the layers?